Common incidences determine eleven full geometry groups

Let $G = \mathrm{HS}$, and let $A = \mathrm{Aut}(G)$, in the acquired faithful models of orders 44 352 000 and 88 704 000. We use the reviewed classification and source coordinates for the coset geometries in D. Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics 33 (2005), 231–236, with the retained supplementary diagrams. The original results remain attributed to that source.

Theorem 1. *The following are the full type-preserving automorphism groups and full correlation groups, respectively. Correlations may permute types.*

Representative	*Source label*	$\mathrm{Aut}(\Gamma)$	$\mathrm{Cor}(\Gamma)$
11	*3.25*	A	A
12	*3.26*	A	$A \times C_2$
23	*3.14*	G	G
26	*3.11*	A	$A \times C_2$
27	*3.10*	A	$A \times C_2$
29	*3.7*	A	A
49	*4.32*	A	A
50	*4.33*	A	A
55	*4.15*	G	$G \times C_2$
71	*4.7*	A	$A \times C_2$
74	*4.3*	A	A

The finite proof uses exact permutation arithmetic and the complete graph automorphism algorithm specified below.

Common-incidence relations. Fix any type a of size $n = 1100$, with base object b and stabilizer P_a. The acquired group is transitive on that type. Let O_k be the orbits of P_a on its objects, of sizes d_k; omit the singleton $\{b\}$. They parametrize the orbits on ordered distinct point pairs, of respective sizes nd_k. For another type j, let $S_j = b^{P_j}$ be the point shadow of its base object. Define e_{jk} to be the number of ordered distinct pairs in S_j belonging to pair orbit k. Then the number of type-j objects incident with both members of a pair in orbit k is

$$m_{jk} = \frac{[G : P_j]e_{jk}}{nd_k}. \tag{1}$$

Indeed, count triples consisting of a type-j object and an ordered pair in its shadow. Counting first by objects gives the numerator; transitivity on the pair orbit makes the count per pair constant. This argument counts objects even when two objects have identical shadows. For $j = a$, the shadow convention is equality within that type, so all off-diagonal counts are zero.

For a distinct pair, form the sorted multiset of pairs $([G : P_j], m_{jk})$ over all types j. This signature is intrinsic: a correlation merely permutes its entries and sends a to a type of the same size. Equal signatures partition the distinct point pairs into intrinsic relations. In representatives 11 and 23 these relations have valencies 280 and 819. In each remaining representative, at every 1100-object anchor, there is exactly one relation of valency 105. Use the valency 280 relation in the first two cases and the valency 105 relation in the others. The literal checks verify symmetry, so each chosen relation is an undirected graph.

Computing the relations in actual coordinates. The saved permutations on 1100 points define a bijective homomorphism from the actual G onto its image E. The specified base point has stabilizer exactly the image of P_a. Recorded elements $t_x \in G$ send that base point to every point x, in order. These facts prove the coset identification $P_a g \mapsto b^g$; the certificate need not assume that a producer's enumeration was a correct transversal. For each ordered pair (x, y) in S_j, its orbit index is that of $y^{t_x^{-1}}$ under E_b. Thus finite counts of these indices compute e_{jk} and hence (1). Transporting the selected base neighbour set by the t_x gives every adjacency row. All specified generators are checked on the entire graph.

Faithfulness of point restriction. We use the elementary shadow lemma from the reviewed common theory: an equivariant encoding of $P_j\backslash G$ is injective precisely when its base value has stabilizer P_j. Begin by encoding an object by its point shadow. If an already identified type k is needed, adjoin the set of encodings of its type-k neighbours. At the base object of type j this is exactly the P_j-orbit of the base type-k encoding. Its cardinality is checked to be $[P_j : P_j \cap P_k]$.

The positive verifier reads the saved encodings in dependency order. It admits an added neighbour set only after type k has been proved injectively encoded, verifies that entire orbit, and intersects the corresponding set stabilizers. The final stabilizer must equal the actual image of P_j. There are sixteen possible 1100-object anchors in these eleven families; every type is identified at every anchor. In particular, representative 49 at its third type needs successive neighbour sets to identify types 1 and 2. No direct-shadow assertion is substituted for these iterations. An automorphism fixing the point type pointwise therefore fixes every object, by induction through the same dependencies. Restriction of the full typed automorphism group to any one anchor is faithful.

The complete upper bound. For each of the sixteen graphs, a complete nauty computation returns order 88 704 000. Every returned generator is checked on every adjacency row. In the literal permutation group D they generate, the verifier checks $|D| = 88\,704\,000$, $E \leq D$ and $D' = E$. Thus E is characteristic in the full graph group. This last equality, rather than agreement of orders alone, supplies the needed invariant copy of the specified group.

Any correlation sends an anchor graph to the graph at its target anchor, by the intrinsic common-incidence definition. If c is such a correlation and $g \in G$, characteristicity implies that the action of cgc^{-1} on the target anchor equals that of some $h \in G$. The typed automorphism $h^{-1}cgc^{-1}$ fixes that anchor pointwise. Faithfulness proves it is the identity. Consequently every full correlation normalizes the specified G.

Exact normalizers and structures. Put $B = \bigcap_i P_i$, and let S, T be the ordered and set-wise stabilizers of the parabolic family in the acquired $\mathrm{Aut}(G)$. The reviewed coset normalizer theorem now applies to the full groups:

$$|\operatorname{Aut}(\Gamma)| = |G : B||S|, \qquad |\operatorname{Cor}(\Gamma)| = |G : B||T|.$$

The same exhaustive transporter calculation tests every type permutation. It verifies actual family images, not just subgroup orders. For the representatives in table order the triples $(|B|, |S|, |T|)$ are

$$(16, 32, 32),\ (16, 32, 64),\ (72, 72, 72),\ (192, 384, 768),$$
$$(192, 384, 768),\ (192, 384, 384),\ (4, 8, 8),\ (4, 8, 8),$$
$$(8, 8, 16),\ (32, 64, 128),\ (64, 128, 128).$$

In every row $S \cap G = B$. The centralizer of G in the full correlation group is the actual quotient $(T \cap G)/B$. Its order is two exactly for representatives $12, 26, 27, 55, 71$, and is otherwise one. In the former cases it is central and meets the typed group trivially; the typed group and this centralizer have full product order. Hence the displayed factors C_2 are direct factors. The

faithful conjugation action identifies the remaining factor with G or $\mathrm{Aut}(G)$ as indicated. This proves all group structures.

Attribution and verification boundary. The coset, shadow and normalizer arguments are reused from the common theory. Formula (1) generalizes the previously reviewed 100-point count; sorted signatures allow arbitrary type interchanges. The present contribution is this application and its explicit finite identifications. The original classification, exact source-coordinate joins, GAP's finite group algorithms and nauty's completeness are acquired premises. Literal small-group controls compare the formula with direct common-neighbour counts, including repeated shadows. No new classification census, historical novelty, conjecture resolution, Lean verification or whole-manuscript completion is asserted here.